UNIT 1 — OPERATING SYSTEM FUNDAMENTALS

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1. Operating System (OS)

Definition:
Operating System is system software that acts as an interface between the user and computer hardware and manages system resources.

Simple Definition:
OS is a manager that controls CPU, Memory, Files and Devices.

Examples:
Windows, Linux, macOS, Android, iOS

Main Functions:
Process Management, Memory Management, File Management, Device Management, Security, Resource Allocation, Error Detection

Interview Answer (20 sec): An Operating System is software that acts as an intermediary between users and hardware while managing resources such as CPU, memory, files and I/O devices.

2. Goals of Operating System

Convenience:
Makes computer easy to use. Without OS, users would directly interact with hardware.

Efficiency:
Utilizes hardware resources effectively. CPU should remain busy.

Ability to Evolve:
Supports new features and hardware. Example: Windows updates.

3. Types of Operating Systems

1. Batch OS:
Jobs are grouped into batches and executed one after another. No user interaction. Advantage: High throughput. Disadvantage: Long turnaround time. Example: Old IBM systems.

2. Multiprogramming OS:
Multiple programs are loaded into memory simultaneously. Goal: Keep CPU busy — if one process waits for I/O, CPU executes another. Advantage: Better CPU utilization.

3. Multitasking OS:
Multiple tasks execute apparently at the same time (e.g. Chrome + Spotify + VS Code). Goal: Improve user experience.

4. Time Sharing OS:
CPU is shared among users using time slices. Interactive, quick response. Example: UNIX, Linux.

5. Multiprocessing OS:
System contains more than one CPU. Advantages: Increased speed, Reliability. Types: Symmetric (SMP, all processors equal), Asymmetric (AMP, one processor controls others).

6. Distributed OS:
Multiple computers appear as a single system. Advantages: Resource sharing, Fault tolerance.

7. Real Time OS (RTOS):
Provides response within a fixed deadline. Hard Real Time — missing deadline = system failure (Air Traffic Control, Missile Systems). Soft Real Time — occasional miss acceptable (Video Streaming, Online Gaming).

Interview Question — Multiprogramming vs Multitasking

Multiprogramming	Multitasking
Multiple programs in memory	Multiple tasks executing
Focus on CPU utilization	Focus on user experience
Mostly system-oriented	Mostly user-oriented

4. Components of Operating System

Kernel:
Core component of OS. Responsibilities: CPU Management, Memory Management, Device Management, Process Scheduling.

Interview Line: Kernel is the heart of the operating system.

Shell:
Interface between user and kernel. CLI (e.g. Bash, CMD) or GUI (e.g. Windows Desktop).
Device Drivers:
Software that enables communication between OS and hardware. Example: Printer Driver.
File System:
Method used by OS to organize files and directories. Examples: FAT32, NTFS, ext4.

5. Kernel Types

Monolithic Kernel:
Entire OS runs in kernel mode. Advantage: Fast. Disadvantages: Less secure, Difficult maintenance. Example: Linux.
Microkernel:
Only essential services run in kernel. Advantages: More secure, Better reliability. Disadvantage: Slower. Example: MINIX.
Hybrid Kernel:
Combination of Monolithic and Microkernel. Examples: Windows, macOS.
Interview Question — Monolithic vs Microkernel

Monolithic	Microkernel
Fast	Secure
Large kernel	Small kernel
Less reliable	More reliable

6. User Mode vs Kernel Mode

User Mode:
Applications run here (Chrome, VS Code, Spotify). Restricted access, cannot directly access hardware.
Kernel Mode:
OS runs here. Full hardware access, privileged mode.
Interview Question — Why User Mode?
Prevents applications from crashing the entire system.

7. System Calls

Definition:
System calls provide communication between user programs and OS.
Interview Definition: A system call is a mechanism through which a user process requests a service from the operating system.
Why Needed?
Applications cannot directly access hardware.
Types of System Calls:
Process Control — fork(), exit(), wait()
File Management — open(), read(), write(), close()
Device Management — request device, release device
Information Maintenance — get time, get date
Communication — send message, receive message
Interview Question — Function Call vs System Call

Function Call	System Call
User space	Kernel space
Faster	Slower
No mode switch	Mode switch required

8. Booting Process

Definition:
Process of loading Operating System into memory.
Steps:

```
Power ON → BIOS/UEFI starts → POST (Power-On Self-Test) →
Bootloader loaded → Kernel loaded → OS starts
```

Types of Booting:
Cold Boot — System starts from OFF state (power button pressed).
Warm Boot — Restart without power off (restart option).

9. BIOS vs UEFI

BIOS	UEFI
Older	Modern
Slower	Faster
Uses MBR	Uses GPT
Supports small disks	Supports large disks

10. Important Placement Concepts

Throughput:
Number of processes completed per unit time. Higher = Better.

Turnaround Time:
Completion Time – Arrival Time.

Response Time:
Time until first response.

Waiting Time:
Time spent in ready queue.

CPU Utilization:
Percentage of time CPU remains busy. Ideal → High.

TOP INTERVIEW QUESTIONS OF UNIT 1

Q1. What is Operating System?
Resource manager and interface between user and hardware.

Q2. What is Kernel?
Core component of OS.

Q3. What is System Call?
Interface between user program and OS.

Q4. Difference between User Mode and Kernel Mode?
User Mode → Restricted. Kernel Mode → Full Access.

Q5. What is Multiprogramming?
Keeping multiple processes in memory to maximize CPU utilization.

Q6. What is Time Sharing?
Sharing CPU among users using time slices.

Q7. What is RTOS?
OS that guarantees response within a deadline.

Q8. What is Booting?
Loading operating system into memory.

Q9. Monolithic vs Microkernel?
Monolithic = Fast. Microkernel = Secure.

Q10. BIOS vs UEFI?
UEFI is modern, faster and supports GPT.

UNIT 2 — PROCESS MANAGEMENT & CPU SCHEDULING

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1. Program vs Process

Program	Process
Passive entity	Active entity
Stored on disk	Executing in memory
Static	Dynamic
No state	Has state

Example:
Program: chrome.exe → When executed: Chrome Process

Interview Answer: A program is a set of instructions stored on disk, whereas a process is a program currently being executed.

2. Process

Definition:
A process is a program in execution.

Components of Process:
Text Section — Program code. Data Section — Global variables. Heap — Dynamic memory allocation. Stack — Function calls and local variables.

Process Memory Layout:

```
Stack (top)
Heap
Data
Code (bottom)
```

3. Process States

New — Process is being created. Ready — Waiting for CPU. Running — Executing on CPU. Waiting/Blocked — Waiting for I/O. Terminated — Execution completed.

```
New → Ready → Running → Terminated
(Running ↔ Waiting ↔ Ready loop while process needs I/O)
```

Interview Question — Ready vs Waiting State
Ready: Waiting for CPU. Waiting: Waiting for I/O resource.

4. Process Control Block (PCB)

Definition:
PCB is a data structure used by OS to store information about a process.

Contains:
Process ID (PID), Process State, Program Counter, CPU Registers, Scheduling Information, Memory Information, I/O Information

Interview Definition: PCB is the identity card of a process.

5. Context Switching

Definition:
Saving state of one process and loading state of another process.

Why Needed?
CPU executes multiple processes. State stored in PCB.

Interview Question — Is Context Switching Useful/Productive?
Useful: Yes. Productive: No — CPU performs switching instead of executing instructions. Therefore, Context Switching = Overhead.

6. CPU Scheduling

Definition:
Process of selecting next process for CPU execution.

Scheduling Goals:
Maximize: CPU Utilization, Throughput. Minimize: Waiting Time, Turnaround Time, Response Time.

Important Formulae:
Turnaround Time = Completion Time – Arrival Time
Waiting Time = Turnaround Time – Burst Time
Response Time = First Response – Arrival Time

7. FCFS Scheduling (First Come First Serve)

Rule:
First process arriving gets CPU first. Type: Non-Preemptive.
Advantages:
Simple, Easy implementation
Disadvantage — Convoy Effect:
Small processes wait behind large process.
Interview Question: FCFS behaves like a FIFO queue.

8. SJF Scheduling (Shortest Job First)

Rule:
Process with shortest burst time executes first. Type: Non-Preemptive.
Advantage:
Minimum Average Waiting Time.
Disadvantage:
Need burst time prediction.
Interview Question — Best Scheduling Algorithm? Theoretically SJF, because it minimizes average waiting time.

9. SRTF Scheduling (Shortest Remaining Time First)

Definition:
Preemptive version of SJF. Process with shortest remaining burst time gets CPU.
Example: If a shorter process arrives, the current process is interrupted.
SJF vs SRTF

SJF	SRTF
Non-preemptive	Preemptive
Cannot interrupt	Can interrupt

10. Priority Scheduling

Rule:
Highest priority process executes first. Types: Preemptive (can interrupt), Non-Preemptive (cannot interrupt).
Problem — Starvation:
Low-priority process waits indefinitely.
Solution — Aging:
Gradually increasing priority of waiting processes.
Interview Question — Starvation vs Deadlock: Starvation = process waits indefinitely. Deadlock = processes wait for each other forever.

11. Round Robin (RR)

Definition:
Each process gets fixed CPU time called Time Quantum.
Example: Time Quantum = 2 ms → P1 → P2 → P3 → P1
Advantages:
Fair, Good response time.
Disadvantage:
Too much context switching.
Interview Question: Very small Time Quantum behaves like Processor Sharing. Very large Time Quantum behaves like FCFS.

12. Multilevel Queue Scheduling

Definition:
Ready queue divided into multiple queues (e.g. System, Interactive, Batch Processes).
Feature:

Processes cannot move between queues.

13. Multilevel Feedback Queue

Definition:
Improved version of Multilevel Queue.

Feature:
Processes can move between queues. Advantage: Most flexible scheduling algorithm.

MLQ vs MLFQ

MLQ	MLFQ
Fixed Queue	Dynamic Queue
No movement	Movement allowed

14. Thread

Definition:
Smallest unit of CPU execution. Also called Lightweight Process.

Process vs Thread

Process	Thread
Heavyweight	Lightweight
Own memory	Shared memory
Slow creation	Fast creation
Expensive switch	Cheap switch

Interview Definition: A thread is the smallest unit of execution inside a process.

Example — Chrome Process: Thread 1 → UI, Thread 2 → Rendering, Thread 3 → Network

15. Benefits of Multithreading

Responsiveness — Application remains responsive.
Resource Sharing — Threads share memory.
Economy — Cheaper than processes.
Scalability — Utilizes multicore CPUs.

16. User Thread vs Kernel Thread

User Thread	Kernel Thread
Managed by user library	Managed by OS
Faster	Slower
Kernel unaware	Kernel aware

17. Threading Models

Many-to-One — Many user threads → One kernel thread. Problem: one blocked thread blocks all.
One-to-One — One user thread → One kernel thread. Most common, used in modern systems.
Many-to-Many — Many user threads → Many kernel threads. Most flexible.

18. fork()

Definition:
Creates a new process. Parent Process = original process. Child Process = newly created process.

Interview Question — What does fork() return? Parent: Child PID. Child: 0. Failure: -1.

19. exec()

Definition:
Replaces current process image with new program.
fork() creates a new process; exec() replaces the process.

fork() vs exec()

fork()	exec()

Creates new process	Replaces process
Child created	No child created

20. Zombie Process

Definition:
Completed process whose PCB still exists. Reason: Parent has not collected exit status.
Memory? No. PID Entry? Yes.
Interview One-Liner: Zombie = Dead process waiting for parent acknowledgment.

21. Orphan Process

Definition:
Child process whose parent has terminated.
What Happens?
OS assigns it to init/system process.
Interview One-Liner: Orphan = Alive child whose parent died.

Zombie vs Orphan

Zombie	Orphan
Dead	Alive
Parent alive	Parent dead
Waiting for cleanup	Adopted by init

UNIT 2 INTERVIEW RAPID FIRE

What is a Process?
Program in execution.
What is PCB?
Data structure containing process information.
What is Context Switching?
Saving one process state and loading another.
Why is Context Switching Overhead?
No useful work is done.
Best Scheduling Algorithm?
SJF (theoretically).
Most Practical Scheduling Algorithm?
Round Robin.
What is Starvation?
Indefinite waiting.
Solution to Starvation?
Aging.
Process vs Thread?
Thread is lightweight and shares memory.
What is fork()?
Creates child process.
What is exec()?
Replaces current process.
What is Zombie Process?
Dead process whose entry still exists.
What is Orphan Process?
Child process whose parent died.

UNIT 3 — PROCESS SYNCHRONIZATION & DEADLOCKS

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1. Process Synchronization

Definition: Coordination of multiple processes/threads so they can safely access shared resources.

Why needed? Simultaneous access to shared data by multiple processes can cause incorrect results.

Example: Two users updating the same bank account balance simultaneously.

2. Shared Resource

Definition: A resource accessible by multiple processes or threads.

Examples: Shared variable, file, database, printer, memory.

3. Race Condition

Definition: Occurs when multiple processes access and modify shared data simultaneously, causing unpredictable results.

Example: Balance=■1000. Thread A withdraws ■100, Thread B withdraws ■200. Expected=■700, but possible result = ■900 or ■800.

Cause: No synchronization. **Solution:** Use Mutex or Semaphore.

4. Critical Section

Definition: The part of a program where shared resources are accessed.

Flow: Entry Section → Critical Section → Exit Section → Remainder Section

Goal: Only one process should execute the critical section at a time.

Requirements of Critical Section

- **Mutual Exclusion** — only one process inside the critical section.
- **Progress** — if critical section is free, a waiting process should eventually enter.
- **Bounded Waiting** — every waiting process gets a fair chance.

Race Condition	Critical Section
Problem	Code section
Incorrect execution	Shared resource access

5. Peterson's Solution

Definition: A software solution for the critical section problem. Works only for 2 processes.

Uses: flag[] and turn variables.

Limitation: Not suitable for modern multicore systems.

6. Mutex (Mutual Exclusion)

Definition: A locking mechanism allowing only one process/thread to access a critical section.

Operations: Lock() and Unlock().

Characteristics: Binary (Locked/Unlocked); only the owner can unlock.

Example flow: Thread A locks → Thread B waits → Thread A unlocks → Thread B locks.

7. Semaphore

Definition: An integer variable used for process synchronization.

Operations: wait() (P operation), signal() (V operation).

Types of Semaphore

- **Binary Semaphore** — value 0 or 1, acts like a mutex.
- **Counting Semaphore** — value 0,1,2,3,... allows multiple processes based on available resources (e.g. printer pool).

Mutex	Semaphore
Locking mechanism	Signaling mechanism
One owner	No owner
Binary only	Binary & Counting
Owner unlocks	Any process can signal

8. Monitor

Definition: A high-level synchronization construct with shared data, procedures, and automatic mutual exclusion.

Advantage: Safer and easier than semaphores.

9. Classical Synchronization Problems

Producer-Consumer Problem

Producer adds data to a buffer; Consumer removes data. Problems: buffer overflow / underflow. Solved using Semaphore + Mutex.

Reader-Writer Problem

Multiple readers can read simultaneously; only one writer can write. Goal: maximize concurrency while maintaining consistency.

Dining Philosophers Problem

Five philosophers need two forks to eat. If everyone picks one fork simultaneously, deadlock occurs.

Solution: Resource ordering, Semaphore, Monitor.

10. Deadlock

Definition: Two or more processes wait indefinitely for resources held by each other.

Example: P1 holds A, waits for B. P2 holds B, waits for A → neither can proceed.

11. Necessary Conditions for Deadlock (Coffman Conditions)

All four must be true for deadlock to occur:

- **Mutual Exclusion** — resource cannot be shared.
- **Hold and Wait** — process holds one resource while waiting for another.
- **No Preemption** — resources cannot be forcibly taken away.
- **Circular Wait** — processes form a cycle of waiting.

Interview trick: If any ONE condition is removed, deadlock cannot occur.

12. Resource Allocation Graph (RAG)

Definition: Graph used to detect possible deadlocks. Nodes = Process, Resource.

Rule: A cycle may indicate deadlock.

13. Deadlock Handling Methods

- **Deadlock Prevention** — break one Coffman condition (e.g. allow resource preemption).
- **Deadlock Avoidance** — allocate resources only if system remains in a Safe State (Banker's Algorithm).
- **Deadlock Detection** — allow deadlock to occur, then detect it.
- **Deadlock Recovery** — terminate process, rollback, or resource preemption.

14. Safe State

Definition: A state where all processes can finish without causing deadlock.

Unsafe State: May lead to deadlock.

Important: Unsafe ≠ Deadlock. Deadlock ⇒ Unsafe. Unsafe ■ Deadlock.

15. Banker's Algorithm

Purpose: Avoid deadlocks by checking if resource allocation keeps the system in a Safe State.

Inputs: Available, Maximum, Allocation, Need.

Formula: Need = Maximum – Allocation

Output: Safe Sequence. Example: P1 → P3 → P4 → P0 → P2

Q: Why called Banker's Algorithm? → Like a banker gives loans only if repayment is possible, OS allocates resources only if all processes can still complete safely.

16. Livelock

Definition: Processes keep changing states but make no progress.

Difference: Deadlock → no movement. Livelock → constant movement, no progress.

17. Deadlock vs Starvation vs Livelock

Deadlock	Starvation	Livelock
Waiting forever	Waiting indefinitely due to scheduling	Active but no progress
Resources blocked	Resource never allocated	Constant retries

18. Busy Waiting (Spin Waiting)

Definition: Process continuously checks a condition instead of sleeping.

Advantage: Fast for very short waits. **Disadvantage:** Wastes CPU cycles.

19. Spinlock

Definition: A lock where a thread repeatedly checks until the lock becomes available.

Use: Multicore systems with short wait times.

TOP INTERVIEW QUESTIONS

Q1. What is Synchronization?

Coordinating processes to safely access shared resources.

Q2. What is a Race Condition?

Incorrect result due to simultaneous access to shared data.

Q3. What is a Critical Section?

Code section accessing shared resources.

Q4. Three requirements of the Critical Section Problem?

Mutual Exclusion, Progress, Bounded Waiting.

Q5. Mutex vs Semaphore?

Mutex is a locking mechanism; Semaphore is a signaling mechanism.

Q6. Binary vs Counting Semaphore?

Binary = 0/1. Counting = multiple resources.

Q7. What is Deadlock?

Processes wait forever for each other's resources.

Q8. Four Necessary Conditions for Deadlock?

Mutual Exclusion, Hold and Wait, No Preemption, Circular Wait.

Q9. What is Banker's Algorithm?

Deadlock avoidance algorithm that checks for a safe state before allocating resources.

Q10. Safe State vs Unsafe State?

Safe state guarantees completion of all processes; unsafe state may lead to deadlock.

Q11. Deadlock vs Starvation?

Deadlock: processes wait for each other. Starvation: a process waits indefinitely because resources are repeatedly given to others.

Q12. Producer-Consumer Problem?

Producer inserts items into a buffer; Consumer removes them. Synchronization prevents overflow and underflow.

UNIT 4 — MEMORY MANAGEMENT

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1. Memory Management

Definition: OS function that allocates, tracks, protects, and deallocates memory for processes.

Objectives: Efficient memory utilization, fast execution, process isolation, protection, sharing.

2. Address Binding

Definition: Mapping a program's logical address to a physical address.

- **Compile-Time** — memory location known before execution.
- **Load-Time** — address determined when program is loaded.
- **Execution-Time** — address translated while running (most common today).

3. Logical vs Physical Address

Logical Address	Physical Address
Generated by CPU	Actual RAM address
Virtual	Real
Seen by Program	Seen by Memory Unit

Interview line: CPU generates logical addresses; MMU translates them into physical addresses.

4. Memory Management Unit (MMU)

Definition: Hardware that converts logical addresses into physical addresses.

Formula: Physical Address = Frame Number + Offset

5. Swapping

Definition: Moving an entire process between RAM and disk to free RAM.

Advantage: Better memory utilization. **Disadvantage:** Slow (disk access).

6. Contiguous Memory Allocation

Each process occupies one continuous block of memory.

Problems: Internal Fragmentation, External Fragmentation.

7. Fragmentation

Internal Fragmentation

Unused memory *inside* an allocated block. Eg: Block=100KB, Process=90KB → Waste=10KB.

External Fragmentation

Free memory exists but scattered in small pieces — enough total memory, but not contiguous.

Internal	External
Inside allocated block	Between allocated blocks
Fixed-size allocation	Variable-size allocation

8. Compaction

Definition: Moving processes so all free memory becomes one large block.

Purpose: Removes external fragmentation. **Disadvantage:** Time-consuming.

9. Paging

Logical memory → Pages; Physical memory → Frames. **Page size = Frame size.**

Address structure: Page Number | Offset

Advantages: Eliminates external fragmentation, efficient memory use.

Disadvantage: Internal fragmentation may occur.

Q: Why Page Size = Frame Size? → So any page fits into any frame without resizing.

10. Page Table

Definition: Maps page numbers to frame numbers, used to translate logical → physical addresses.

11. Translation Lookaside Buffer (TLB)

Definition: Small, high-speed cache storing recent page table entries — reduces translation time.

- **TLB Hit** — page found in TLB → fast.
- **TLB Miss** — not found → OS checks page table → slower.

12. Segmentation

Memory divided into logical segments: Code, Data, Stack, Heap.

Advantages: Matches program structure; easy sharing & protection.

Disadvantage: External fragmentation.

13. Paging vs Segmentation

Paging	Segmentation
Fixed-size pages	Variable-size segments
Programmer invisible	Programmer visible
Internal fragmentation	External fragmentation

14. Virtual Memory

Definition: Lets programs larger than RAM run by using disk as extension of memory.

Advantages: Larger address space, better multitasking, efficient memory usage.

Don't say "Virtual Memory is RAM." It is a TECHNIQUE that extends memory using disk.

15. Demand Paging

Pages loaded into memory only when needed → saves memory, improves efficiency.

16. Page Fault

Definition: Occurs when a required page is not present in RAM.

Steps: CPU requests page → not in RAM → OS loads from disk → page table updated → execution resumes.

Q: Is Page Fault an error? → No, it's a normal event handled by the OS.

17. Page Replacement Algorithms

Used when RAM is full and OS must choose a page to replace.

FIFO (First In First Out)

Replace oldest page. Simple, but may replace frequently used pages.

LRU (Least Recently Used)

Replace page unused for longest time. Good practical performance, higher implementation cost.

Optimal Page Replacement

Replace page that won't be used for longest time in future. Minimum faults, but impossible in practice (future unknown).

Clock (Second Chance)

FIFO + reference bit giving recently used pages a second chance. Efficient LRU approximation.

Algorithm	Practical?	Performance
FIFO	Yes	Moderate
LRU	Yes	Very Good
Clock	Yes	Good
Optimal	No	Best (Theoretical)

18. Belady's Anomaly

Definition: Increasing number of page frames results in MORE page faults.

Happens in: FIFO. **Does NOT happen in:** LRU, Optimal.

Interview favorite: Which algorithm suffers from Belady's Anomaly? → FIFO

19. Thrashing

Definition: System spends more time swapping pages than executing processes.

Cause: Too many page faults. **Effect:** CPU utilization decreases, system slows.

Solution: Increase RAM, reduce degree of multiprogramming, use better page replacement algorithms.

20. Local vs Global Page Replacement

Local	Global
Replace pages of same process	Replace any process's pages
Stable	Better overall utilization

21. Memory Protection

Definition: Prevents one process from accessing another process's memory.

Methods: Base Register, Limit Register, Paging Protection, Segmentation Protection.

22. Cache Memory vs Main Memory

Cache	RAM

Very Fast	Slower
Small	Large
Expensive	Cheaper
CPU accesses first	Used if cache misses

TOP INTERVIEW QUESTIONS

Q1. What is Memory Management?

Managing allocation and deallocation of memory.

Q2. Logical vs Physical Address?

Logical generated by CPU; Physical is actual RAM address.

Q3. What is Paging?

Fixed-size pages mapped to fixed-size frames.

Q4. What is Segmentation?

Memory divided into logical variable-sized segments.

Q5. Paging vs Segmentation?

Paging = fixed-size pages; Segmentation = variable-size logical segments.

Q6. What is Virtual Memory?

Using disk as an extension of RAM.

Q7. What is Demand Paging?

Load pages only when needed.

Q8. What is a Page Fault?

Required page not present in RAM.

Q9. What is TLB?

Cache storing recent page table entries to speed up translation.

Q10. Internal vs External Fragmentation?

Internal = waste inside block; External = free memory scattered in small blocks.

Q11. What is Thrashing?

Excessive paging causing poor performance.

Q12. Best Page Replacement Algorithm?

Theoretical: Optimal. Practical: LRU (Clock = efficient approximation).

Q13. Which Algorithm Shows Belady's Anomaly?

FIFO.

UNIT 5 — FILE SYSTEMS, DISK MANAGEMENT & IPC

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1. File System

Definition: The method used by an OS to organize, store, retrieve, and manage files on storage devices.

Objectives: Organize files, fast access, efficient storage, security, reliability.

Examples: FAT32, NTFS, ext4, APFS.

2. File

Definition: A collection of related data stored on secondary storage.

Attributes: Name, Identifier, Size, Type, Location, Owner, Permissions, Creation Time, Modification Time.

3. File Operations

- Create
- Open
- Read
- Write
- Append
- Seek
- Rename
- Delete
- Close

4. File Access Methods

Sequential Access

Data accessed one after another. Example: Music player reading an audio file.

Direct (Random) Access

Any record can be accessed directly. Example: Database systems.

Indexed Access

Uses an index to locate records quickly. Example: Book index.

Sequential	Direct	Indexed
One by one	Any position	Through index
Simple	Fast	Fastest search

5. Directory Structure

- **Single-Level Directory** — one directory for all files; problem: name conflicts.
- **Two-Level Directory** — separate directory for each user.
- **Tree Directory** — hierarchical structure, most common.
- **Acyclic Graph Directory** — allows shared files, no cycles.

6. File Allocation Methods

Contiguous Allocation

Files occupy consecutive disk blocks. **Advantages:** fast access, simple. **Disadvantages:** external fragmentation, difficult file expansion.

Linked Allocation

Each block points to the next block. **Advantage:** no external fragmentation. **Disadvantage:** slow random access.

Indexed Allocation

Index block stores addresses of all file blocks. **Advantage:** fast random access. **Disadvantage:** index block overhead.

Contiguous	Linked	Indexed
Fast	Flexible	Random access
External fragmentation	No fragmentation	Index overhead

7. Free Space Management

Methods used to keep track of free disk blocks:

- **Bitmap** — each bit represents one block.
- **Linked List** — free blocks linked together.
- **Grouping** — stores addresses of free blocks in groups.
- **Counting** — stores starting block and count.

8. Hard Link vs Soft Link

Hard Link: Points directly to file data. Deleting original file does not delete data.

Soft (Symbolic) Link: Points to file name/path. Deleting original file breaks the link.

Hard Link	Soft Link
Same inode	Different inode
Survives original deletion	Breaks if original deleted
Cannot span file systems	Can span file systems

9. Disk Scheduling

Definition: Technique used by the OS to decide the order of disk I/O requests.

Goals: Reduce seek time, improve throughput, reduce response time.

10. FCFS Disk Scheduling

Rule: Serve requests in arrival order. **Advantages:** simple, fair. **Disadvantage:** high seek time.

11. SSTF (Shortest Seek Time First)

Rule: Choose the request closest to current head position. **Advantage:** lower seek time. **Disadvantage:** starvation possible.

12. SCAN (Elevator Algorithm)

Rule: Disk head moves in one direction servicing requests, then reverses. **Advantage:** fair and efficient.

13. C-SCAN (Circular SCAN)

Rule: Head moves in one direction only; after reaching the end, jumps back to the beginning. **Advantage:** uniform waiting time.

14. LOOK

Rule: Similar to SCAN, but head reverses at the last pending request instead of the disk end.

15. C-LOOK

Rule: Circular version of LOOK. Head jumps back after servicing the last request.

Algorithm	Advantage	Drawback
FCFS	Simple	High seek time
SSTF	Low seek time	Starvation
SCAN	Fair	Extra movement
C-SCAN	Uniform wait	Jump overhead
LOOK	Less movement	Slightly complex
C-LOOK	Efficient	More complex

16. RAID (Redundant Array of Independent Disks)

Definition: Technique that combines multiple disks for performance and/or fault tolerance.

- **RAID 0** — Striping; fastest; no fault tolerance.
- **RAID 1** — Mirroring; high reliability; storage overhead.
- **RAID 5** — Striping + Distributed Parity; good balance of speed and reliability.
- **RAID 10 (1+0)** — Striping + Mirroring; high performance and fault tolerance; expensive.

Interview favorite: RAID 0 has no fault tolerance. RAID 1 uses mirroring.

17. Inter-Process Communication (IPC)

Definition: IPC allows processes to exchange data and synchronize with each other.

IPC Methods

- **Pipe** — communication between related processes; half-duplex.
- **Named Pipe (FIFO)** — communication between unrelated processes.
- **Message Queue** — processes exchange messages through the OS.
- **Shared Memory** — processes share the same memory region; fastest IPC method; requires synchronization.
- **Socket** — communication between processes on same or different machines; used in client-server apps.
- **Signal** — notification sent to a process to indicate an event (e.g. Ctrl+C interrupt).

IPC Method	Speed	Related Processes
Pipe	Fast	Yes
Named Pipe	Fast	No
Message Queue	Moderate	Yes/No
Shared Memory	Fastest	Yes/No
Socket	Network	Yes/No

Signal	Notification	Yes/No

18. Daemon Process

Definition: A background process that provides system services.

Examples: Web server, print service.

19. Copy-on-Write (COW)

Definition: Parent and child initially share the same memory after fork(). Actual copying occurs only when one process modifies the memory.

Advantages: Faster fork(), saves memory.

20. Memory-Mapped File

Definition: Maps a file directly into a process's virtual memory.

Advantages: Faster file access, efficient sharing.

21. Linux Commands (Interview Basics)

Command	Purpose
pwd	Present working directory
ls	List files
cd	Change directory
mkdir	Create directory
rm	Remove file
cp	Copy file
mv	Move/Rename file
cat	Display file contents
grep	Search text
find	Find files
ps	List running processes
top	System monitoring
kill	Terminate process
chmod	Change permissions
df	Disk usage
free	Memory usage

TOP INTERVIEW QUESTIONS

Q1. What is a File System?

A method used by the OS to organize and manage files.

Q2. Sequential vs Direct Access?

Sequential reads data in order; Direct accesses any location.

Q3. Contiguous vs Linked vs Indexed Allocation?

Contiguous is fast but fragments; Linked avoids fragmentation but has slow random access; Indexed supports efficient random access with index overhead.

Q4. Hard Link vs Soft Link?

Hard links point to the same inode; soft links point to the file path.

Q5. Best Disk Scheduling Algorithm?

No universally best algorithm — SSTF minimizes seek time but may cause starvation; SCAN/LOOK provide a good balance and are commonly used.

Q6. What is RAID?

A technique that combines multiple disks for better performance and/or reliability.

Q7. Fastest IPC Method?

Shared Memory (requires synchronization).

Q8. Pipe vs Socket?

Pipe is mainly for local inter-process communication; Socket supports communication over networks as well.

Q9. What is a Daemon Process?

A background service process that runs without user interaction.

Q10. What is Copy-on-Write?

Memory pages are copied only when modified after fork().

UNIT 1 — COMPUTER NETWORK FUNDAMENTALS

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1. Computer Network

Definition: A collection of interconnected devices that communicate and share data and resources using communication links and protocols.

Simple explanation: A network allows computers and devices to exchange information and share resources like files, printers, and internet connections.

Examples: Internet, College Wi-Fi, Office LAN, Mobile Network.

Why Do We Need Computer Networks?

- Resource Sharing
- File Sharing
- Communication
- Internet Access
- Remote Access
- Cost Reduction
- Distributed Computing

Advantages

- Fast Communication
- Resource Sharing
- Centralized Data Management
- Scalability
- Reliability
- Easy Backup

Disadvantages

- Security Risks
- Virus/Malware Spread
- Setup Cost
- Maintenance Required

Interview definition: A computer network is a group of interconnected devices that communicate using standardized protocols to exchange data and share resources.

2. Components of a Computer Network

- **Sender** — device that sends data.
- **Receiver** — device that receives data.
- **Transmission Medium** — path through which data travels (e.g. Fiber Optic, Wi-Fi, Ethernet Cable).
- **Message** — actual data being transmitted.
- **Protocol** — rules governing communication (e.g. TCP/IP).

3. Types of Networks

PAN (Personal Area Network)

Small network around a person. Range: ~10 meters. Example: Bluetooth between phone and earbuds.

LAN (Local Area Network)

Network within a small area. Examples: Home Wi-Fi, School Lab, Office. Advantages: high speed, low cost.

MAN (Metropolitan Area Network)

Network covering a city. Example: city-wide cable TV network.

WAN (Wide Area Network)

Network spanning countries or continents. Example: Internet.

WLAN (Wireless LAN)

LAN using Wi-Fi instead of cables.

VPN (Virtual Private Network)

Creates a secure, encrypted connection over a public network. Use: remote office access.

Network	Coverage
PAN	Few meters
LAN	Building
MAN	City
WAN	Country/World

4. Network Topologies

Bus Topology

All devices connected to one backbone cable. **Advantages:** cheap, easy installation. **Disadvantages:** backbone failure stops network; difficult fault detection.

Star Topology

All devices connected to a central switch/hub. **Advantages:** easy maintenance, easy fault isolation. **Disadvantage:** central device failure stops network. Most common — used in modern Ethernet networks.

Ring Topology

Each device connected to two neighbors. **Advantage:** orderly data transmission. **Disadvantage:** one break can disrupt communication.

Mesh Topology

Every device connected to every other device. **Advantages:** highly reliable, fault tolerant. **Disadvantages:** expensive, complex.

Tree Topology

Combination of star topologies in a hierarchy.

Hybrid Topology

Combination of two or more topologies.

Topology	Advantage	Disadvantage
Bus	Cheap	Backbone failure
Star	Easy maintenance	Central device failure
Ring	Predictable transmission	Single break affects network

Mesh	Reliable	Expensive
Tree	Scalable	Complex
Hybrid	Flexible	Costly

5. Network Devices

Repeater

Regenerates weak signals. OSI Layer: Physical Layer (Layer 1).

Hub

Broadcasts incoming data to all connected devices. Features: no intelligence, no MAC table, more collisions. OSI Layer: Physical Layer (Layer 1).

Bridge

Connects two LAN segments and filters traffic using MAC addresses. OSI Layer: Data Link Layer (Layer 2).

Switch

Forwards data only to the intended destination using MAC addresses. Features: MAC address table, fewer collisions, better performance. OSI Layer: Data Link Layer (Layer 2).

Router

Connects different networks using IP addresses. Uses: internet access, path selection. OSI Layer: Network Layer (Layer 3).

Gateway

Connects networks using different protocols. Acts as a protocol converter.

Modem

Converts digital signals to analog and vice versa. Example: home broadband modem.

Access Point (AP)

Provides wireless connectivity to devices. Example: Wi-Fi access point in offices.

Most Asked Interview Question — Hub vs Switch

Hub	Switch
Broadcasts to all devices	Sends only to destination
Slower	Faster
More collisions	Fewer collisions
Layer 1	Layer 2

Switch vs Router

Switch	Router
Uses MAC address	Uses IP address
Connects devices in same network	Connects different networks
Layer 2	Layer 3

Bridge vs Switch

Bridge	Switch
Few ports	Many ports
Slower	Faster
Older technology	Modern technology

6. Transmission Media

Guided Media (Wired)

- **Twisted Pair Cable** — cheapest, common in Ethernet.
- **Coaxial Cable** — better shielding, used in cable TV.
- **Optical Fiber** — uses light, highest speed, long-distance communication, immune to electromagnetic interference.

Unguided Media (Wireless)

Examples: Wi-Fi, Bluetooth, Radio Waves, Infrared, Satellite.

Medium	Speed	Cost
Twisted Pair	Moderate	Low
Coaxial	Good	Moderate
Optical Fiber	Very High	High
Wireless	Varies	Moderate

7. Data Transmission Modes

Simplex

One-way communication. Example: Keyboard → Computer.

Half Duplex

Both directions, but one at a time. Example: Walkie-Talkie.

Full Duplex

Both directions simultaneously. Example: Phone Call.

Simplex	Half Duplex	Full Duplex
One-way	Two-way (one at a time)	Two-way (simultaneous)

8. Switching Techniques

Circuit Switching

Dedicated communication path established before data transfer. Example: traditional telephone system. **Advantages:** reliable, constant bandwidth. **Disadvantages:** resource wastage, connection setup time.

Packet Switching

Data divided into packets, each may take a different route. Example: Internet. **Advantages:** efficient, better bandwidth utilization. **Disadvantage:** variable delay.

Message Switching

Entire message stored and forwarded at each intermediate node. **Disadvantage:** high delay.

Circuit	Packet	Message
Dedicated path	Packets	Whole message
Fast after setup	Efficient	Slow
Telephone	Internet	Legacy systems

9. Network Performance Metrics

Bandwidth

Maximum amount of data that can be transmitted per second. Unit: bps (bits per second).

Throughput

Actual amount of data successfully transmitted. Key point: Throughput ≤ Bandwidth.

Latency

Time taken for data to travel from source to destination. Lower is better.

Delay

Total time required for packet delivery. Includes: Processing Delay, Queuing Delay, Transmission Delay, Propagation Delay.

Jitter

Variation in packet delay. Important for: video calls, online gaming, VoIP.

Metric	Meaning
Bandwidth	Maximum capacity
Throughput	Actual transfer rate
Latency	Travel time
Delay	Total packet delivery time
Jitter	Variation in delay

10. Collision & Broadcast Domain

Collision Domain

Area where packet collisions can occur. Hub: one collision domain. Switch: one collision domain per port.

Broadcast Domain

Area where broadcast packets are received. Routers separate broadcast domains.

TOP INTERVIEW QUESTIONS

Q1. What is a Computer Network?

A collection of interconnected devices that communicate using protocols.

Q2. LAN vs WAN?

LAN covers a small area with high speed; WAN covers large geographical areas and connects multiple LANs.

Q3. Hub vs Switch?

Hub broadcasts to all devices; Switch forwards data only to the intended destination using MAC addresses.

Q4. Switch vs Router?

Switch connects devices within the same network using MAC addresses; Router connects different networks using IP addresses.

Q5. Why is a Switch better than a Hub?

Because it reduces unnecessary traffic and minimizes collisions.

Q6. Circuit Switching vs Packet Switching?

Circuit switching uses a dedicated path; Packet switching divides data into packets that may take different routes.

Q7. Bandwidth vs Throughput?

Bandwidth is the maximum possible capacity; Throughput is the actual achieved data transfer rate.

Q8. Simplex vs Half Duplex vs Full Duplex?

Simplex: one-way. Half Duplex: two-way, one direction at a time. Full Duplex: two-way simultaneously.

Q9. Why is Optical Fiber preferred over Copper?

Higher bandwidth, longer distance, lower signal loss, and immunity to electromagnetic interference.

Q10. What is a Collision Domain?

A network segment where simultaneous transmissions can collide.

UNIT 2 — OSI MODEL, TCP/IP MODEL & NETWORK PROTOCOLS

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1. Protocol

Definition: A set of rules that governs communication between devices on a network.

Why needed? Without protocols, devices would not understand how to send, receive, or interpret data.

Examples: HTTP, HTTPS, TCP, UDP, IP, FTP, DNS.

Interview definition: A protocol is a standardized set of rules that enables communication between network devices.

2. OSI Model

Definition: The OSI (Open Systems Interconnection) Model is a 7-layer reference model that standardizes network communication.

Purpose: Standardize networking, simplify troubleshooting, ensure interoperability between vendors.

OSI Layers (Top → Bottom)

Layer	Name	PDU
7	Application	Data
6	Presentation	Data
5	Session	Data
4	Transport	Segment
3	Network	Packet
2	Data Link	Frame
1	Physical	Bits

Memory trick: All People Seem To Need Data Processing (Application → Physical)

3. Layer 7 – Application Layer

Function: Provides network services directly to user applications.

Protocols: HTTP, HTTPS, FTP, SMTP, DNS, DHCP, Telnet, SSH.

Examples: Web Browser, Gmail, WhatsApp.

4. Layer 6 – Presentation Layer

Function: Encryption, Decryption, Compression, Data Translation.

Example: HTTPS encryption.

5. Layer 5 – Session Layer

Function: Establishes, maintains, and terminates a session.

Example: Video conferencing session.

6. Layer 4 – Transport Layer

Function: Reliable delivery, flow control, error recovery, end-to-end communication.

Protocols: TCP, UDP.

7. Layer 3 – Network Layer

Function: Logical addressing, routing, path selection.

Protocols: IP, ICMP, ARP (often considered between Layer 2 and 3).

Device: Router.

8. Layer 2 – Data Link Layer

Function: MAC addressing, framing, error detection, media access control.

Device: Switch.

9. Layer 1 – Physical Layer

Function: Transmits raw bits over the medium.

Devices: Hub, Repeater.

Interview Favorite — Devices and OSI Layers

Device	Layer
Repeater	Layer 1
Hub	Layer 1
Bridge	Layer 2
Switch	Layer 2
Router	Layer 3
Gateway	Multiple Layers

10. Encapsulation

Definition: Process of adding headers (and sometimes trailers) as data moves down the OSI stack.

Flow: Application Data → Transport Header → Network Header → Data Link Header → Bits

11. Decapsulation

Definition: Removing headers as data moves up the receiving device's protocol stack.

Why Encapsulation? → Allows each layer to add its own information independently.

12. TCP/IP Model

Definition: The practical networking model used on the Internet.

TCP/IP	OSI Equivalent
Application	Application + Presentation + Session
Transport	Transport
Internet	Network
Network Access	Data Link + Physical

OSI	TCP/IP

7 Layers	4 Layers
Reference Model	Practical Model
Theoretical	Internet Standard

Which model is actually used on the Internet? → TCP/IP Model

13. HTTP

Full Form: HyperText Transfer Protocol.

Features: Application Layer, Stateless, used for websites.

Default Port: 80

14. HTTPS

Full Form: HyperText Transfer Protocol Secure.

Difference: HTTP + SSL/TLS Encryption.

Default Port: 443

HTTP	HTTPS
Not encrypted	Encrypted
Port 80	Port 443
Less secure	More secure

Why is HTTPS more secure? → It encrypts communication using SSL/TLS, protecting data from interception.

15. FTP

Full Form: File Transfer Protocol.

Purpose: Transfers files between computers.

Default Ports: 21 (Control), 20 (Data).

16. SMTP

Full Form: Simple Mail Transfer Protocol.

Purpose: Sending emails. **Port:** 25 (traditional).

17. POP3

Full Form: Post Office Protocol Version 3.

Purpose: Downloads emails to the local device. **Port:** 110

18. IMAP

Full Form: Internet Message Access Protocol.

Purpose: Keeps emails on the server and synchronizes across devices. **Port:** 143

POP3 vs IMAP

POP3	IMAP
Downloads mail	Syncs mail
Single device	Multiple devices

Server copy may be removed	Server copy retained

19. DNS

Full Form: Domain Name System.

Definition: Translates domain names into IP addresses. Example: google.com → 142.250.x.x

Interview definition: DNS is the Internet's phonebook.

DNS Resolution Process

Browser → Local DNS Cache → ISP DNS Server → Root Server → TLD Server (.com) → Authoritative DNS Server → IP Address Returned

20. DHCP

Full Form: Dynamic Host Configuration Protocol.

Purpose: Automatically assigns IP Address, Subnet Mask, Default Gateway, DNS Server.

DORA Process

Discover → Offer → Request → Acknowledge

Without DHCP? → User must manually configure IP addresses.

21. Telnet

Purpose: Remote login. **Port:** 23. **Security:** Not encrypted.

22. SSH

Full Form: Secure Shell.

Purpose: Secure remote login. **Port:** 22

Telnet vs SSH

Telnet	SSH
No encryption	Encrypted
Less secure	Secure

23. SNMP

Full Form: Simple Network Management Protocol.

Purpose: Monitoring network devices like routers and switches.

24. Common Port Numbers

Protocol	Port
HTTP	80
HTTPS	443
FTP	21
SSH	22
Telnet	23

SMTP	25
DNS	53
DHCP	67/68
POP3	110
IMAP	143

25. Socket

Definition: An endpoint for communication between two processes over a network.

Socket = IP Address + Port Number. Example: 192.168.1.10 : 8080

26. URL vs URI

URL	URI
Identifies and locates a resource	Identifies a resource
Every URL is a URI	Not every URI is a URL

27. API (Interview Basics)

Definition: An Application Programming Interface (API) allows software applications to communicate with each other.

Example: A mobile app calling a weather service over HTTP.

Note: APIs commonly use HTTP/HTTPS, but an API is not the same thing as HTTP.

TOP INTERVIEW QUESTIONS

Q1. What is a Protocol?

A set of rules for communication between network devices.

Q2. How many layers are there in the OSI Model?

Seven.

Q3. Which model is used in real networks?

TCP/IP Model.

Q4. Which layer does routing?

Network Layer (Layer 3).

Q5. Which layer does encryption?

Presentation Layer (Layer 6).

Q6. Which layer does switching?

Data Link Layer (Layer 2).

Q7. What is DNS?

Maps domain names to IP addresses.

Q8. What is DHCP?

Automatically assigns network configuration such as IP addresses.

Q9. HTTP vs HTTPS?

HTTPS uses SSL/TLS encryption; HTTP does not.

Q10. POP3 vs IMAP?

POP3 downloads emails locally; IMAP synchronizes emails across devices.

Q11. Telnet vs SSH?

SSH is encrypted and secure; Telnet is not.

Q12. What is a Socket?

An endpoint for network communication identified by an IP address and port number.

UNIT 3 — IP ADDRESSING, SUBNETTING & ROUTING

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1. IP Address

Definition: A unique logical address assigned to every device connected to a network.

Purpose: Identifies a device, enables communication, helps routers forward packets.

Example: 192.168.1.10

2. IPv4

Definition: Internet Protocol Version 4. **Size:** 32 bits.

Format: 192.168.1.10. **Total Addresses:** $2^{32} \approx 4.3$ Billion.

Advantages: Simple, widely supported. **Disadvantages:** Address exhaustion.

3. IPv6

Definition: Internet Protocol Version 6. **Size:** 128 bits.

Example: 2001:0db8:85a3:0000:0000:8a2e:0370:7334

Advantages: Huge address space, better security support, no address shortage.

IPv4 vs IPv6

IPv4	IPv6
32-bit	128-bit
Decimal	Hexadecimal
~4.3 Billion addresses	Extremely large address space
NAT often required	NAT generally not needed
Example: 192.168.1.1	Example: 2001:db8::1

4. IP Address Structure

Every IPv4 address has two parts: **Network ID** (identifies the network) and **Host ID** (identifies the device).

Example: 192.168.1.10 → Network ID = 192.168.1, Host ID = 10

5. IP Address Classes

Class	Range	Default Mask
A	1 – 126	255.0.0.0
B	128 – 191	255.255.0.0
C	192 – 223	255.255.255.0
D	224 – 239	Multicast
E	240 – 255	Experimental

Memory trick: A→1, B→128, C→192, D→224, E→240

6. Private vs Public IP

Private IP

Used inside local networks. Cannot be accessed directly from the Internet.

Ranges: 10.0.0.0/8 | 172.16.0.0 – 172.31.255.255 | 192.168.0.0/16

Public IP

Assigned by Internet Service Provider (ISP). Accessible over the Internet.

Private IP	Public IP
Internal use	Internet use
Not globally unique	Globally unique
Free	Assigned by ISP

7. Static IP vs Dynamic IP

Static IP

Never changes. Used for servers, CCTV, web hosting.

Dynamic IP

Automatically assigned by DHCP. Used for home users, mobile devices.

Static	Dynamic
Fixed	Changes
Manual	Automatic
Server	Home users

8. Subnet Mask

Definition: Separates the Network ID and Host ID. Example: 255.255.255.0

Common Masks

CIDR	Subnet Mask
/8	255.0.0.0
/16	255.255.0.0
/24	255.255.255.0

9. CIDR (Classless Inter-Domain Routing)

Definition: Method of allocating IP addresses using prefix length instead of classes.

Example: 192.168.1.0/24 → 24 bits = Network, 8 bits = Host

Advantages: Efficient IP allocation, smaller routing tables, reduces address wastage.

10. Subnetting (Interview Level)

Definition: Dividing one large network into smaller subnetworks.

Why? Better security, reduced congestion, efficient IP utilization.

Example: 192.168.1.0/24 → 192.168.1.0/25 and 192.168.1.128/25

Why Subnetting? → To improve network efficiency and reduce broadcast traffic.

11. NAT (Network Address Translation)

Definition: Technique that converts private IP addresses into public IP addresses.

Why needed? IPv4 addresses are limited.

Flow: Private IP → Router (NAT) → Public IP → Internet

Advantages: Saves public IP addresses, improves security.

12. Default Gateway

Definition: The router through which devices communicate with other networks.

Interview definition: Gateway to leave the local network.

13. MAC Address

Definition: Unique physical address assigned to a Network Interface Card (NIC).

Size: 48 bits. **Example:** 00:1A:2B:3C:4D:5E

MAC vs IP

MAC Address	IP Address
Physical	Logical
Permanent	Can change
Layer 2	Layer 3

14. ARP

Full Form: Address Resolution Protocol.

Purpose: Converts IP Address → MAC Address.

15. RARP

Full Form: Reverse Address Resolution Protocol.

Purpose: Converts MAC Address → IP Address.

Note: RARP is obsolete and has largely been replaced by BOOTP and DHCP.

ARP vs RARP

ARP	RARP
IP → MAC	MAC → IP
Commonly used	Obsolete

16. ICMP

Full Form: Internet Control Message Protocol.

Purpose: Reports network errors and diagnostic information.

Used by: ping, traceroute (along with other mechanisms).

Why ping works? → Because it uses ICMP Echo Request and Echo Reply.

17. Routing

Definition: Process of forwarding packets from source to destination, performed by a router.

18. Static Routing

Definition: Routes are manually configured.

Advantages: Simple, predictable. **Disadvantage:** Difficult to manage in large networks.

19. Dynamic Routing

Definition: Routes are learned automatically.

Advantages: Scalable, automatic updates.

Static	Dynamic
Manual	Automatic
Small networks	Large networks

20. Routing Algorithms

Distance Vector

Example: RIP. Features: shares routing table with neighbors; simpler but slower convergence.

Link State

Example: OSPF. Features: builds a complete network topology; faster convergence; more scalable.

Distance Vector	Link State
RIP	OSPF
Simpler	More complex
Slower convergence	Faster convergence

21. Routing Table

Definition: A table maintained by routers containing the best path to different networks.

Contains: Destination Network, Next Hop, Metric, Interface.

22. Broadcast Address

Definition: Address used to send data to all devices in a subnet. Example: 192.168.1.255

23. Loopback Address

Definition: Special IP used to test the local network stack. Example: 127.0.0.1

What is localhost? → 127.0.0.1

24. Unicast, Broadcast & Multicast

- **Unicast** — one sender → one receiver.
- **Broadcast** — one sender → all devices in the subnet.
- **Multicast** — one sender → selected group of receivers.

Type	Communication
Unicast	One to One
Broadcast	One to All

Multicast	One to Many (Group)

TOP INTERVIEW QUESTIONS

Q1. What is an IP Address?

A logical address used to uniquely identify a device on a network.

Q2. IPv4 vs IPv6?

IPv4 uses 32-bit addresses; IPv6 uses 128-bit addresses with a much larger address space.

Q3. What is a MAC Address?

A unique physical address assigned to a network interface card.

Q4. MAC vs IP?

MAC identifies hardware at Layer 2; IP identifies devices logically at Layer 3.

Q5. What is ARP?

Converts IP Address into MAC Address.

Q6. What is NAT?

Converts private IP addresses to public IP addresses.

Q7. Why is NAT used?

To conserve IPv4 addresses and allow multiple devices to share one public IP.

Q8. What is DHCP?

Automatically assigns IP addresses and other network configuration.

Q9. What is Subnetting?

Dividing a network into smaller subnetworks to improve efficiency and reduce broadcast traffic.

Q10. Static vs Dynamic Routing?

Static routes are manually configured; Dynamic routes are learned automatically using routing protocols.

Q11. What is ICMP?

A protocol used for error reporting and diagnostics (e.g., ping).

Q12. What is the Loopback Address?

127.0.0.1 (localhost).

UNIT 4 — TRANSPORT LAYER, TCP, UDP & RELIABLE DATA TRANSFER

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1. Transport Layer

Definition:
The Transport Layer (Layer 4) provides end-to-end communication between applications running on different devices.

Responsibilities:

- Process-to-process communication
- Reliable delivery
- Flow control
- Error recovery
- Segmentation & Reassembly
- Multiplexing

Protocols:
TCP, UDP

2. TCP (Transmission Control Protocol)

Definition:
TCP is a connection-oriented and reliable transport layer protocol.

Features:

- Reliable
- Connection-oriented
- Error detection
- Flow control
- Congestion control
- Ordered delivery

Used By:
HTTP, HTTPS, FTP, SMTP, SSH

3. UDP (User Datagram Protocol)

Definition:
UDP is a connectionless transport layer protocol.

Features:

- Fast
- No connection setup
- No reliability guarantee
- No ordering
- Low overhead

Used By:
DNS, VoIP, Video Streaming, Online Gaming, Live Broadcasting

TCP vs UDP

TCP	UDP
Connection-oriented	Connectionless
Reliable	Unreliable
Ordered delivery	No ordering
Error recovery	No recovery
Slower	Faster
Larger header	Smaller header

Interview Question — Why is UDP used for video calls?
Because speed is more important than perfect reliability. A few lost packets are preferable to delays caused by retransmissions.

4. Connection-Oriented vs Connectionless

Connection-Oriented:
Connection established before data transfer. Example: TCP

Connectionless:
No connection setup. Example: UDP

Connection-Oriented	Connectionless
Reliable	Fast
TCP	UDP

5. TCP Three-Way Handshake

Purpose:
Establishes a reliable TCP connection.

Steps:

```
Client Server
SYN ---------------------->
<------------------- SYN+ACK
ACK ---------------------->
Connection Established
```

Steps:
1) Client sends SYN. 2) Server replies with SYN+ACK. 3) Client sends ACK — connection established.

Interview Definition:
Three-way handshake establishes synchronization between client and server before communication begins.

Why Three Steps?
To ensure both client and server are ready and agree on initial sequence numbers.

6. TCP Four-Way Handshake (Connection Termination)

Purpose:
Gracefully closes a TCP connection.

```
Client Server
FIN ---------------------->
<---------------------- ACK
<---------------------- FIN
ACK ---------------------->
```

Steps:
1) Client sends FIN. 2) Server acknowledges. 3) Server sends FIN. 4) Client acknowledges — connection closed.

Interview Question — Why four steps instead of three?
Because TCP connections are full-duplex, each direction is closed independently.

7. Sequence Number

Definition:
Unique number assigned to every TCP byte for ordered delivery.

Purpose:

- Detect missing data
- Maintain order
- Retransmission

8. Acknowledgment Number

Definition:
Indicates the next byte expected by the receiver.

9. Flow Control

Definition:
Controls the sender's transmission speed so the receiver is not overwhelmed.

TCP Uses:
Sliding Window.

Example:
Fast sender → Slow receiver → Flow control slows sender.

10. Sliding Window Protocol

Definition:
Allows multiple packets to be sent before waiting for acknowledgments.

Advantages:

- Better bandwidth utilization
- Higher throughput

Interview Question — Why Sliding Window?
To improve efficiency by avoiding unnecessary waiting after every packet.

11. Congestion Control

Definition:
Prevents excessive traffic in the network.

Goal:
Avoid network congestion.

TCP Congestion Algorithms:

- Slow Start
- Congestion Avoidance
- Fast Retransmit
- Fast Recovery

Interview Tip: You don't usually need to explain the algorithms in detail unless specifically asked. Know their names and purpose.

Flow Control vs Congestion Control

Flow Control	Congestion Control
Protects receiver	Protects network
Receiver based	Network based

12. Error Detection

Purpose:
Detect corrupted data.

Checksum — Definition:
Mathematical value used to detect errors. Used by: TCP, UDP, IP

CRC (Cyclic Redundancy Check) — Definition:
Advanced error detection method using polynomial division. Used in: Ethernet, Data Link Layer

Checksum vs CRC

Checksum	CRC
Simpler	More accurate
Transport Layer	Data Link Layer

13. Reliable Data Transfer

TCP ensures reliability using:

- Sequence Numbers
- Acknowledgments
- Retransmissions
- Checksums

14. Multiplexing

Definition:
Multiple applications share the same network connection.

Example:
Browser, Music, Video Call — all use the network simultaneously.

15. Demultiplexing

Definition:
Receiver delivers incoming packets to the correct application using port numbers.

16. Port Number

Definition:
A logical communication endpoint used to identify applications.

Range:

Range	Purpose
0–1023	Well-known ports
1024–49151	Registered ports
49152–65535	Dynamic (Ephemeral) ports

Interview Tip: Server listens on a well-known port (e.g., 80 for HTTP). Client typically uses an ephemeral (temporary) port chosen by the operating system.

Common Ports

Service	Port
HTTP	80
HTTPS	443
FTP	21
SSH	22
DNS	53
SMTP	25

17. Socket

Definition:
A socket is an endpoint used for communication between two processes.

Formula:

```
Socket = IP Address + Port Number
```

Example:

```
192.168.1.5 : 8080
```

18. MSS (Maximum Segment Size)

Definition:
The largest amount of TCP payload that can be sent in one TCP segment.

Purpose:
Improves efficiency by avoiding fragmentation.

19. MTU (Maximum Transmission Unit)

Definition:
The maximum size of a complete packet that can be transmitted over a network link.

Ethernet MTU:
1500 bytes

MTU vs MSS

MTU	MSS
Entire packet size	TCP payload only
Network Layer concept	Transport Layer concept

20. Retransmission

Definition:
Lost packets are sent again after timeout or duplicate acknowledgments.

21. RTT (Round Trip Time)

Definition:
Time taken for a packet to travel to the destination and back.

Used to:

- Calculate timeout

- Improve TCP performance

22. QoS (Quality of Service)

Definition:
Technique used to prioritize important network traffic.

Examples:
Video Calls, Online Meetings, Voice Calls

TOP INTERVIEW QUESTIONS

Q1. TCP vs UDP?
TCP is reliable and connection-oriented. UDP is faster and connectionless.

Q2. Why is UDP used for gaming?
Lower latency is preferred over guaranteed delivery.

Q3. What is Three-Way Handshake?
TCP connection establishment using: SYN → SYN-ACK → ACK

Q4. Why Four-Way Handshake?
TCP closes each communication direction independently.

Q5. What is Flow Control?
Prevents the sender from overwhelming the receiver.

Q6. Flow Control vs Congestion Control?
Flow Control protects the receiver. Congestion Control protects the network.

Q7. What is Sliding Window?
Allows multiple packets to be transmitted before receiving acknowledgments.

Q8. What is a Port Number?
A logical identifier for a network application or service.

Q9. What is a Socket?
An endpoint identified by an IP address and a port number.

Q10. Checksum vs CRC?
Checksum is simpler and commonly used at the Transport Layer. CRC is stronger and commonly used at the Data Link Layer.

Q11. What is MSS?
Maximum TCP payload size in one segment.

Q12. What is MTU?
Maximum packet size that can be transmitted over a network link.

UNIT 5 — APPLICATION LAYER, NETWORK SECURITY & FINAL PLACEMENT REVISION

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1. Application Layer

Definition:
The Application Layer is the top layer of the network stack that provides network services directly to user applications.

Examples:
Web Browser, Email Client, WhatsApp, Chrome, Outlook

Common Protocols:
HTTP, HTTPS, DNS, DHCP, FTP, SMTP, POP3, IMAP, SSH

2. DNS (Domain Name System)

Definition:
Converts domain names into IP addresses.

Example:

```
www.google.com → 142.250.xxx.xxx
```

Why Needed?
Humans remember names, computers communicate using IP addresses.

DNS Resolution Process:

```
Browser → Browser Cache → OS Cache → Local DNS Resolver →
Root Server → TLD Server (.com) → Authoritative DNS Server →
IP Address Returned
```

Interview Question — What happens if DNS fails?
The website cannot be reached using its domain name, though accessing it directly via its IP address may still work (if permitted).

3. DHCP

Definition:
Automatically assigns: IP Address, Subnet Mask, Gateway, DNS Server.

DORA Process:

```
Discover → Offer → Request → Acknowledge
```

4. HTTP

Full Form:
HyperText Transfer Protocol

Features:
Stateless, Application Layer, Uses TCP, Default Port 80

5. HTTPS

Definition:
HTTP secured using SSL/TLS encryption.

Port:
443

Advantages:
Confidentiality, Integrity, Authentication

HTTP vs HTTPS

HTTP	HTTPS
Port 80	Port 443
Not encrypted	Encrypted
Less secure	Secure

6. SSL & TLS

SSL (Secure Sockets Layer):

Older encryption protocol.

TLS (Transport Layer Security):

Modern replacement for SSL. Used in HTTPS.

Purpose:

Encryption, Authentication, Data Integrity

Interview Tip: Although people often say "SSL certificate," modern HTTPS actually uses TLS.

7. Digital Certificate

Definition:

Electronic document that verifies the identity of a website. Issued by Certificate Authority (CA).

Contains:

Domain Name, Public Key, Expiry Date, CA Signature

8. FTP

Definition:

Transfers files between computers.

Ports:

21 → Control, 20 → Data

9. Email Protocols

SMTP:

Sending emails. Port 25

POP3:

Downloads emails. Port 110

IMAP:

Synchronizes emails across devices. Port 143

SMTP	POP3	IMAP
Send	Download	Sync

10. Cookies

Definition:

Small files stored by browsers to remember user information.

Uses:

Login sessions, Preferences, Shopping Cart

11. Cache

Definition:

Temporary storage used to improve performance.

Browser Cache stores:

Images, CSS, JavaScript — Purpose: Faster website loading.

12. Proxy Server

Definition:

An intermediary server between client and destination server.

Uses:

Content filtering, Caching, Privacy

13. Reverse Proxy

Definition:

Receives client requests and forwards them to backend servers.

Uses:

Load balancing, Security, SSL termination

Proxy vs Reverse Proxy

Proxy	Reverse Proxy
Represents client	Represents server

Hides client	Hides servers

14. Firewall

Definition:
Monitors and filters incoming and outgoing network traffic.

Types:
Hardware Firewall, Software Firewall

Purpose:
Prevent unauthorized access.

15. VPN (Virtual Private Network)

Definition:
Creates an encrypted tunnel over a public network.

Advantages:
Privacy, Secure remote access, Data encryption

VPN vs Proxy

VPN	Proxy
Encrypts all traffic	Usually forwards application traffic only
More secure	Less secure
Entire device	Individual application/browser

16. CDN (Content Delivery Network)

Definition:
A network of distributed servers that delivers content from the nearest location.

Benefits:
Faster websites, Lower latency, Reduced server load

Example:
Netflix, YouTube, Amazon

17. Load Balancer

Definition:
Distributes incoming requests among multiple servers.

Benefits:
High availability, Scalability, Fault tolerance

Example:

```
Users → Load Balancer → Server 1 / Server 2 / Server 3
```

18. Socket Programming (Basics)

Socket:
Endpoint of communication. Formula: Socket = IP + Port

Client:
Initiates communication.

Server:
Accepts communication.

19. Linux Networking Commands

Command	Purpose
ping	Connectivity (uses ICMP)
traceroute / tracert	Shows the path packets take
ip / ifconfig	Displays or configures IP addresses
netstat	Displays active network connections
nslookup	Finds DNS information
arp	Displays ARP table

route / ip route	Displays routing table
curl	Transfers data using URLs; tests REST APIs
wget	Downloads files from the Internet

20. CORS (Interview Basics)

Full Form:
Cross-Origin Resource Sharing

Definition:
A browser security mechanism that allows or restricts requests from one origin to another.

Why Needed?
To safely allow frontend applications to access resources hosted on different domains.

21. Same-Origin Policy (SOP)

Definition:
Browser security policy that prevents scripts from accessing resources from a different origin unless explicitly allowed.

SOP vs CORS

SOP	CORS
Restricts cross-origin access	Allows controlled cross-origin access

22. Complete Browser Request Flow (Most Asked Interview Question)

```
User types www.google.com → DNS Resolution → IP Address Found →
TCP Three-Way Handshake → TLS Handshake (HTTPS) → HTTP Request Sent →
Server Processes Request → HTTP Response Returned → Browser Renders Page
```

Interview Tip: This is one of the most frequently asked Computer Networks interview questions. Practice explaining each step clearly.

TOP INTERVIEW QUESTIONS

Q1. What is DNS?
Maps domain names to IP addresses.

Q2. What is DHCP?
Automatically assigns IP addresses and network configuration.

Q3. HTTP vs HTTPS?
HTTPS is HTTP secured using TLS encryption.

Q4. SSL vs TLS?
TLS is the modern successor to SSL.

Q5. What is a Digital Certificate?
Verifies the identity of a website and contains its public key.

Q6. What is a Firewall?
Filters network traffic to prevent unauthorized access.

Q7. What is a VPN?
Creates a secure encrypted connection over a public network.

Q8. Proxy vs Reverse Proxy?
Proxy represents clients; Reverse Proxy represents servers.

Q9. What is a CDN?
A distributed network of servers that delivers content from locations closer to users.

Q10. What is a Load Balancer?
Distributes client requests across multiple servers.

Q11. What is CORS?
A mechanism that allows controlled cross-origin requests.

Q12. Explain what happens when you enter www.google.com in a browser.
DNS lookup → TCP handshake → TLS handshake → HTTP request → Server response → Browser renders the webpage.

DBMS UNIT 1 — DATABASE FUNDAMENTALS & DATABASE DESIGN

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1. What is Data?

Definition:
Data is a collection of raw facts and figures.

Example:
Rudransh, 21, Delhi — these values alone have little meaning.

2. What is Information?

Definition:
Information is processed and meaningful data.

Example:
Rudransh is 21 years old and lives in Delhi.

Data vs Information

Data	Information
Raw facts	Processed data
No meaning	Meaningful
Input	Output

3. Database

Definition:
A Database is an organized collection of related data stored electronically for easy access, retrieval, and management.

Examples:
Student Database, Banking Database, Hospital Database, Library Database

4. What is DBMS?

Definition:
A Database Management System (DBMS) is software used to create, store, retrieve, update, and manage databases efficiently.

Examples:
MySQL, PostgreSQL, Oracle Database, Microsoft SQL Server, SQLite

Responsibilities of DBMS:
Store, Retrieve, Update, Delete data; Security; Backup; Concurrency Control; Recovery

Interview Definition: A DBMS is software that acts as an interface between users/applications and the database, enabling efficient and secure data management.

5. Why Do We Need a DBMS?

Without a DBMS:
Data redundancy, Data inconsistency, Poor security, Difficult backup, Difficult sharing, No concurrency control

Advantages:
Reduced redundancy, Better consistency, Data security, Data sharing, Backup & recovery, Concurrency control, Data integrity

Disadvantages:
Expensive, Complex, Requires skilled administrators, Performance overhead for small applications

6. File System vs DBMS

File System	DBMS
Stores files	Stores structured data
High redundancy	Low redundancy
Poor security	Better security
Difficult backup	Easy backup
No concurrency	Supports concurrency

No recovery	Recovery mechanisms available

Interview Question — Why is DBMS better than File System?
Because DBMS reduces redundancy, improves security, supports concurrent users, maintains data integrity, and provides backup and recovery.

7. What is RDBMS?

Definition:
An RDBMS (Relational Database Management System) stores data in the form of tables (relations).

Features:
Tables, Rows, Columns, Primary Keys, Foreign Keys, Relationships

Examples:
MySQL, PostgreSQL, Oracle, SQL Server

DBMS vs RDBMS

DBMS	RDBMS
May not use tables	Uses tables
Relationships optional	Relationships supported
Smaller applications	Large applications
Less strict	Follows relational model

8. Types of DBMS (Interview Overview)

Hierarchical DBMS — Tree structure.

Network DBMS — Graph structure.

Relational DBMS (Most Common) — Stores data in tables.

Object-Oriented DBMS — Stores objects instead of only tables.

Interview Tip: Most modern applications use RDBMS.

9. Three-Schema Architecture

Purpose:
Provides data abstraction.

External Level:
What each user sees. Example: Customer sees only account details.

Conceptual Level:
Overall database structure. Contains: Tables, Relationships, Constraints

Internal Level:
Physical storage of data. Contains: Files, Indexes, Storage details

```
Users → External Level → Conceptual Level → Internal Level → Database
```

10. Data Independence

Definition:
Ability to modify one level without affecting another.

Physical Data Independence:
Changes in storage do NOT affect logical structure. Example: Changing indexing.

Logical Data Independence:
Changes in logical schema do NOT affect user views. Example: Adding a new column.

Physical	Logical
Storage changes	Schema changes
Easier	Harder

11. Schema vs Instance

Schema:
Blueprint or structure of the database. Rarely changes.

Instance:
Actual data stored at a particular time. Changes frequently.

Example:

```
Schema: Student(ID, Name, Branch)
Instance: 101 Rahul CSE | 102 Amit IT
```

Schema	Instance
Structure	Actual data
Stable	Changes frequently

12. ER Model (Entity Relationship Model)

Definition:
A conceptual model used to design databases.
Components:
Entity, Attribute, Relationship

13. Entity

Definition:
A real-world object that can be identified uniquely.
Examples:
Student, Employee, Product, Customer

14. Entity Set

Collection of similar entities. Example: All students in a college.

15. Attributes

Definition:
Properties describing an entity. Example — Student: Roll No, Name, Branch, Age
Types of Attributes:
Simple — cannot be divided (e.g. Age)
Composite — can be divided (e.g. Address → City, State, PIN Code)
Single-Valued — one value (e.g. Date of Birth)
Multi-Valued — multiple values (e.g. Phone Numbers)
Derived — calculated from another attribute (e.g. Age derived from Date of Birth)

16. Relationship

Definition:
Association between entities.

```
Student → Enrolls → Course
```

Relationship Degree:

Degree	Example
Unary	Employee manages Employee
Binary	Student enrolls Course
Ternary	Doctor prescribes Medicine to Patient

17. Cardinality

Definition:
Defines how many entities can participate in a relationship.
One-to-One (1:1) — One person ↔ One passport
One-to-Many (1:M) — One department → Many employees
Many-to-One (M:1) — Many students → One department
Many-to-Many (M:N) — Many students ↔ Many courses

18. Participation

Total Participation:
Every entity must participate.
Partial Participation:
Participation is optional.

19. Weak Entity & Strong Entity

Weak Entity:
Entity that cannot exist without another (owner) entity. Example: Dependent of an Employee.

Strong Entity:
Has its own primary key. Can exist independently. Example: Student.

Strong	Weak
Independent	Depends on another entity
Own primary key	Partial key

20. ER Diagram Symbols (Must Remember)

Symbol	Meaning
Rectangle	Entity
Double Rectangle	Weak Entity
Oval	Attribute
Double Oval	Multi-valued Attribute
Diamond	Relationship
Underlined Attribute	Primary Key

21. Primary Introduction to Tables

Relation — A table.
Tuple — A row.
Attribute — A column.
Domain — Allowed values for an attribute.

TOP INTERVIEW QUESTIONS

Q1. What is DBMS?
Software that manages databases efficiently.

Q2. Database vs DBMS?
Database stores data. DBMS manages the database.

Q3. File System vs DBMS?
DBMS provides better security, reduced redundancy, concurrency control, and recovery.

Q4. DBMS vs RDBMS?
RDBMS stores data in tables and supports relationships using keys.

Q5. What is Data Independence?
Ability to modify one level of the database without affecting another.

Q6. Schema vs Instance?
Schema is the database structure; Instance is the current data.

Q7. What is an Entity?
A real-world object that can be uniquely identified.

Q8. What is Cardinality?
Defines the number of entities participating in a relationship.

Q9. Weak Entity vs Strong Entity?
Weak entities depend on another entity; Strong entities have their own primary key.

Q10. What is an ER Model?
A conceptual model used to design a database by representing entities, attributes, and relationships.

DBMS UNIT 2 — KEYS, CONSTRAINTS, NORMALIZATION & TRANSACTIONS

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1. Relational Model

Definition:
A Relational Model stores data in the form of tables (relations).

Basic Terminology:

Term	Meaning
Relation	Table
Tuple	Row
Attribute	Column
Domain	Allowed values
Degree	Number of columns
Cardinality	Number of rows

2. Keys

What is a Key?
A Key is an attribute or a set of attributes used to identify records uniquely.

Super Key:
Any attribute or combination of attributes that uniquely identifies a row. Example: StudentID, or StudentID+Name.

Candidate Key:
A minimal Super Key — uniquely identifies a record with no unnecessary attributes. Example: StudentID, Email.

Primary Key:
The Candidate Key chosen to uniquely identify every record. Properties: Unique, Cannot be NULL, One per table. Example: StudentID.

Alternate Key:
Candidate Keys not selected as Primary Key.

Composite Key:
Key formed using multiple attributes. Example: StudentID + CourseID.

Foreign Key:
An attribute that refers to the Primary Key of another table. Purpose: Creates relationships between tables.

Key Comparison

Key	Purpose
Super Key	Unique identification
Candidate Key	Minimal Super Key
Primary Key	Selected Candidate Key
Alternate Key	Remaining Candidate Keys
Composite Key	Multiple columns
Foreign Key	Creates relationships

3. Constraints

Definition:
Rules that ensure data accuracy and consistency.

Constraint	Purpose
NOT NULL	No NULL values
UNIQUE	No duplicates
PRIMARY KEY	Unique identifier (Unique + NOT NULL)
FOREIGN KEY	Maintains relationship between tables

CHECK	Validate values, e.g. Age > 18
DEFAULT	Assigns default value, e.g. Country = India

4. Integrity Constraints

Entity Integrity:
Primary Key cannot be NULL.

Referential Integrity:
Foreign Key must reference an existing Primary Key.

Interview Question — Why Referential Integrity?
Prevents orphan records and keeps relationships valid.

5. Functional Dependency (FD)

Definition:
If one attribute determines another attribute, it is called a Functional Dependency. Notation: A → B (A determines B).

Example:
StudentID → Name — one StudentID always has one Name.

Why Important?
Functional Dependencies are the basis of Normalization.

6. Normalization

Definition:
Process of organizing data to reduce redundancy and avoid anomalies.

Advantages:
Removes duplicate data, Saves storage, Improves consistency, Easier maintenance

Disadvantages:
More tables, More joins

7. Anomalies

Insertion Anomaly:
Cannot insert data without unnecessary information.

Update Anomaly:
Need to update the same information in multiple places.

Deletion Anomaly:
Deleting one record accidentally removes useful information.

8. First Normal Form (1NF)

Rule:
Atomic values only, No repeating groups, No multi-valued attributes.

Example:
■ Phone: 9876, 9999 → ■ Separate rows or related table.

9. Second Normal Form (2NF)

Rule:
Must be in 1NF, and remove Partial Dependency.

Partial Dependency:
Occurs when a non-key attribute depends on only part of a composite key.

10. Third Normal Form (3NF)

Rule:
Must be in 2NF, and remove Transitive Dependency.

Transitive Dependency:
A non-key attribute depends on another non-key attribute.

Example:
StudentID → DepartmentID → DepartmentName. DepartmentName should be moved to a separate table.

11. BCNF (Boyce-Codd Normal Form)

Rule:
Every determinant must be a Candidate Key.

Interview Tip: For placements, know 1NF, 2NF, 3NF well. BCNF is usually asked only as a concept.

Normalization Comparison

Normal Form	Removes
1NF	Repeating groups
2NF	Partial Dependency
3NF	Transitive Dependency
BCNF	Candidate Key violations

12. Denormalization

Definition:
Combining tables to improve read performance.

Advantage:
Faster queries.

Disadvantage:
Data redundancy increases.

13. Transactions

Definition:
A Transaction is a sequence of operations executed as a single logical unit.

Example — Bank Transfer:
Withdraw ■100 → Deposit ■100. Both operations must succeed together.

14. Transaction States

```
Active → Partially Committed → Committed
If failure occurs → Failed → Aborted
```

States:
Active, Partially Committed, Committed, Failed, Aborted

15. ACID Properties

Atomicity:
Either all operations happen or none.

Consistency:
Database remains valid before and after the transaction.

Isolation:
Concurrent transactions do not interfere.

Durability:
Committed changes remain even after a crash.

16. Commit & Rollback

Commit:
Permanently saves changes.

Rollback:
Undoes uncommitted changes.

Commit	Rollback
Save	Undo

17. Concurrency Control

Definition:
Allows multiple users to access data safely at the same time.

Why Needed?
Prevent data inconsistency.

18. Problems in Concurrent Transactions

Dirty Read:
Reading uncommitted data.

Non-Repeatable Read:
Same row gives different values in one transaction because another transaction updated it.

Phantom Read:
A repeated query returns extra or missing rows because another transaction inserted or deleted matching records.

Problem	Meaning
Dirty Read	Reads uncommitted data
Non-Repeatable Read	Same row changes
Phantom Read	Result set changes

19. Locking

Shared Lock (S):
Read only. Multiple transactions can hold it simultaneously.

Exclusive Lock (X):
Read and write. Only one transaction can hold it.

Shared	Exclusive
Read	Read + Write
Multiple allowed	Only one allowed

20. Two-Phase Locking (2PL)

Definition:
Protocol that ensures conflict serializability.

Phases:
Growing Phase — Acquire locks only. Shrinking Phase — Release locks only.

Interview Tip: Know the concept. No need to memorize algorithms.

21. Deadlock

Definition:
A situation where two or more transactions wait forever for each other to release resources.

Necessary Conditions:
1) Mutual Exclusion 2) Hold and Wait 3) No Preemption 4) Circular Wait

Handling Deadlock:
Prevention, Detection, Recovery

Deadlock vs Starvation

Deadlock	Starvation
Everyone waits	One waits indefinitely
Circular dependency	Resource never allocated

TOP INTERVIEW QUESTIONS

Q1. What is a Primary Key?
Uniquely identifies each record and cannot be NULL.

Q2. Primary Key vs Foreign Key?
Primary Key identifies records in its own table; Foreign Key references the Primary Key of another table.

Q3. Candidate Key vs Super Key?
Candidate Key is a minimal Super Key.

Q4. What is Normalization?
Organizing data to reduce redundancy and improve consistency.

Q5. Explain 1NF, 2NF, and 3NF.
1NF → Atomic values. 2NF → Remove partial dependency. 3NF → Remove transitive dependency.

Q6. What are ACID Properties?
Atomicity, Consistency, Isolation, Durability.

Q7. What is a Transaction?

A sequence of operations executed as one logical unit.

Q8. Commit vs Rollback?

Commit saves changes; Rollback undoes uncommitted changes.

Q9. What is a Deadlock?

Two or more transactions wait indefinitely for each other to release resources.

Q10. Shared Lock vs Exclusive Lock?

Shared locks allow multiple readers; Exclusive locks allow a single reader/writer with write access.

DBMS UNIT 3 — INDEXING, SQL CONCEPTS, RECOVERY & INTERVIEW QUESTIONS

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1. What is an Index?

Definition:
An Index is a data structure that improves the speed of data retrieval from a database. (Think of it like an index in a book.)

Without Index:
Searching checks every row (Full Table Scan) → Slow.

With Index:
Index points directly to the record → Fast.

Advantages:
Faster searching, Faster sorting, Faster joins, Improves query performance

Disadvantages:
Uses extra storage, Slower INSERT/UPDATE/DELETE

Interview Question — Why not index every column?
Because every index consumes storage and must be updated whenever data changes, which slows write operations.

2. Types of Index

Clustered Index:
Determines the physical order of data in the table. Only one per table. Example: Student table sorted by StudentID.

Non-Clustered Index:
Stores pointers to the actual data without changing its physical order. Multiple allowed.

Clustered vs Non-Clustered Index

Clustered Index	Non-Clustered Index
Changes physical order	Does not change physical order
One per table	Multiple allowed
Faster range queries	Good for selective lookups

3. Joins

Definition:
A Join combines rows from two or more tables based on a related column.

Inner Join:
Returns only matching records.

Left Join:
All rows from Left Table + Matching rows from Right Table (NULL if no match).

Right Join:
All rows from Right Table + Matching rows from Left Table.

Full Outer Join:
All matching rows + all non-matching rows.

Self Join:
Table joins with itself. Example: Employee → Manager relationship.

Cross Join:
Every row from Table A joins with every row from Table B (Cartesian Product).

Join Comparison

Join	Returns
Inner	Matching rows
Left	All Left + Matching Right
Right	All Right + Matching Left
Full	All rows
Self	Same table
Cross	Cartesian Product

4. View

Definition:
A View is a virtual table created using the result of a SQL query.

Advantages:
Security, Simplicity, Reusable queries

Example:
HR can view only Employee Name and Salary, not personal details.

View vs Table

View	Table
Virtual	Physical
Stores query	Stores data
Less storage	Stores actual records

5. Materialized View

Definition:
A view that stores the query result physically.

View	Materialized View
Calculates every time	Stores result
Slower	Faster reads
Less storage	Uses storage

Interview Tip: Normal View = Virtual. Materialized View = Physical Copy.

6. Trigger

Definition:
A Trigger is a procedure that executes automatically when an event occurs (INSERT, UPDATE, DELETE).

Uses:
Audit logging, Validation, Automatic updates

Example:
Whenever salary changes, log old salary automatically.

7. Stored Procedure

Definition:
A Stored Procedure is a precompiled collection of SQL statements stored in the database.

Advantages:
Faster execution, Reusable, Better security, Reduced network traffic

Trigger vs Stored Procedure

Trigger	Stored Procedure
Runs automatically	Called manually
Event-driven	User/Application driven

8. Cursor (Interview Basics)

Definition:
A Cursor processes query results one row at a time.

Why Used?
When row-by-row processing is required.

Interview Tip: Cursors are slower than set-based operations and should be avoided when possible.

9. Recovery

Definition:
Recovery restores the database after failures.

Types of Failure:
System Crash, Power Failure, Disk Failure, Software Error

10. Log-Based Recovery

Definition:
Every database operation is recorded in a log file before execution.

Purpose:
If the database crashes, replay or undo operations using logs.

11. Checkpoint

Definition:
A point where all modified data is safely written to disk.

Advantages:
Faster recovery, Smaller log processing

Backup vs Recovery

Backup	Recovery
Copy of data	Restore data
Prevents data loss	Restores after failure

12. OLTP vs OLAP

OLTP (Online Transaction Processing):
Used for Banking, Shopping, ATM, Ticket Booking. Characteristics: Many small transactions, Fast updates, Current data.

OLAP (Online Analytical Processing):
Used for Reports, Business Intelligence, Analytics. Characteristics: Large queries, Historical data, Decision making.

OLTP	OLAP
Transactions	Analysis
Current data	Historical data
Fast updates	Complex queries

13. Database Replication (Basic)

Definition:
Copying data from one database server to another.

Advantages:
High availability, Backup, Load distribution

14. Database Sharding (Basic)

Definition:
Splitting a large database into smaller parts called shards.

Why Needed?
Large-scale applications. Example: Users A–M → Server 1, Users N–Z → Server 2.

Replication vs Sharding

Replication	Sharding
Copies data	Splits data
Improves availability	Improves scalability

15. CAP Theorem (Basic)

Definition:
In a distributed database, only two of the following three can be fully guaranteed simultaneously: Consistency, Availability, Partition Tolerance.

Interview Tip: Know only the concept. Deep details are rarely asked in campus interviews.

16. NoSQL (Interview Overview)

Definition:
Databases that do not use the traditional relational table model.

Examples:
MongoDB, Cassandra, Redis

SQL vs NoSQL

SQL	NoSQL
Tables	Documents / Key-Value / Graph
Fixed Schema	Flexible Schema
Vertical Scaling	Horizontal Scaling

17. Most Asked DBMS Comparisons

Comparison	Key Difference
DBMS vs RDBMS	Tables & Relationships
Primary vs Foreign Key	Identification vs Reference
Candidate vs Super Key	Minimal vs Any Unique Combination
View vs Table	Virtual vs Physical
View vs Materialized View	Virtual vs Stored
Trigger vs Stored Procedure	Automatic vs Manual
Clustered vs Non-Clustered Index	Physical vs Logical
OLTP vs OLAP	Transactions vs Analytics
Replication vs Sharding	Copy vs Split

TOP INTERVIEW QUESTIONS

Q1. What is an Index?
A data structure that speeds up data retrieval.

Q2. Why use Indexing?
To reduce query execution time.

Q3. Clustered vs Non-Clustered Index?
Clustered changes the physical order of data; Non-clustered stores pointers without changing data order.

Q4. What is a View?
A virtual table created from a SQL query.

Q5. Trigger vs Stored Procedure?
Trigger executes automatically on database events; Stored Procedure is called explicitly.

Q6. What is Recovery?
The process of restoring a database after a failure.

Q7. What is a Checkpoint?
A point where modified data is flushed to disk to speed up recovery.

Q8. OLTP vs OLAP?
OLTP handles day-to-day transactions; OLAP supports analysis and reporting.

Q9. What is Database Replication?
Maintaining copies of the same database on multiple servers.

Q10. What is Sharding?
Splitting a database into smaller parts to improve scalability.

Q11. SQL vs NoSQL?
SQL databases use tables and fixed schemas; NoSQL databases support flexible data models.

Q12. Why not create an Index on every column?
Indexes improve reads but consume storage and slow down INSERT, UPDATE, and DELETE operations.